

# MARA DUMITRU

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## EDUCATION

### Minerva University (1% acceptance rate) – San Francisco, CA

Aug 2023 – May 2027

*Bachelor of Science, Data Science; GPA: 3.78/4.0*

- Relevant Coursework: Machine Learning, Data Structures and Algorithms, Bayesian Statistics, Optimization Methods, Linear Algebra, Probability and Statistics
- TA: Probability and Statistics (selected from top 7% of class)

## EXPERIENCE

### AI Intern – Blue Ocean Initiative

Sep 2024 – Aug 2025

*Remote*

- Improved forecasting accuracy 53% using optimized XGBoost models across 2,490 NASA time-series datasets
- Achieved 0.85+ SSIM and 0.006 MAE for 5-day Pacific forecasts with SA-ConvLSTM hybrid model in PyTorch
- Communicated findings to 130+ researchers from 5+ universities by building Matplotlib/Tableau visualization pipeline identifying 1,000+ pollution hotspots for non-technical stakeholders

### Software Engineering Intern – Seoul Nooks

Jan 2025 – Jul 2025

*Seoul, Korea*

- Engineered Django/PostgreSQL CRM with 25+ models processing \$70K+ bookings
- Built multi-tier pricing engine for 500+ day stays with 0.01% accuracy and automated payments saving \$3K/year
- Architected Python/SQL pipelines reducing admin workload 50% and removing 80% of external dependencies; integrated REST APIs and deployed on Render

### Computer Science Curriculum Intern – Minerva University

Aug 2023 – Aug 2024

*San Francisco, CA*

- Improved student test scores by 45% teaching Python, NumPy, and Pandas to 50+ students across 80 sessions
- Reduced administrative workload 80% by engineering automated allocation system processing 70+ records
- Grew study group attendance 4x by A/B testing 20+ message variants using data-driven engagement strategies; earned 2 promotions for teaching excellence

## PROJECTS

### Capillary-Driven Microfluidic Network

- Extending Lab-on-a-Chip ML-GA framework with Prof. Duriez to optimize multi-inlet networks; applicable to cell biology and bioprocessing

### AI Taxonomy

- Led team of 4 building automated NLP classification system with 26 LLM agent prompts; reduced regulatory document analysis from 2+ hours to 8 minutes across 10 countries; deployed on Vercel

### 1st Prize AI Consensus: HERMEX

- Built real-time AI tutoring platform in 24-hour hackathon; owned frontend and JSON integration for bidirectional audio streaming across 6+ conversation states

### AI Consensus Circuit Tracing

- Built attribution pipeline mapping feature circuits across Gemma-2/Llama-3/Qwen3; created interactive visualization framework for AI safety research

### Anthropic Event Organizer

- Co-organized the 1st Claude AI event in Buenos Aires for 70+ attendees as 1 of 5 organizers; managed venue logistics, led bilingual communications with local partners, and delivered live technical demo

## SKILLS

- **Languages:** Python, SQL (PostgreSQL, MySQL), JavaScript, Bash, LaTeX
- **ML/AI:** PyTorch, TensorFlow, XGBoost, Scikit-Learn, LLMs (GPT, Llama, Gemma), Time-Series Neural Networks, Hybrid Modeling, NLP
- **Data Science:** NumPy, Pandas, PyMC, Statistical Modeling, Time Series, Feature Engineering, A/B Testing
- **Frameworks & Tools:** Django, React, OpenAI API, Gradio, Optuna, Wandb, Matplotlib, Tableau, Jupyter
- **Cloud & Deployment:** AWS (S3, Lambda), Git, CI/CD, Vercel